Recall our stiff semilinear problem

$$u_t = Lu + f(u)$$

Another approach to combine implicit and explicit integration is that of ImEx methods. Here's a simple one:

$$u^{n+1} = u^n + \underbrace{hf(u^n)}_{\text{Euler for } f} + \underbrace{\frac{h}{2}\left(Lu^n + Lu^{n+1}\right)}_{\text{Trapezoidal for } L}$$

This method is of course only 1st-order accurate. More generally we can consider ImEx Runge-Kutta methods:

$$Y_i = u_n + h\sum_{j=1}^{i} a_{ij} L(Y_j) + h\sum_{j=1}^{i-1} \hat{a}_{ij} f(Y_j)$$

$$u_{n+1} = u_n + h\sum_{j=1}^{s} b_j L(Y_j) + h\sum_{j=1}^{s} \hat{b}_j f(Y_j)$$

$$\begin{array}{c|c|c|c} c & A & \hat{A} & \hat{c} \\ \hline & b & \hat{b} & \end{array}$$

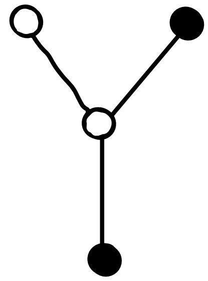
Accuracy

The separate schemes (A, b, c) and $(\hat{A}, \hat{b}, \hat{c})$ must each satisfy the usual order conditions. Additionally, they must satisfy combined conditions like

$$\sum_i b_i \hat{c}_i = \frac{1}{2} \qquad \sum_i \hat{b}_i c_i = \frac{1}{2}$$

It's common to take either $b = \hat{b}$ or $c = \hat{c}$ in order to simplify things. The full set of OCs can be associated with bicolored rooted trees.

For example:



$$\sum_{i,j,k,\ell} b_i \hat{a}_{ij} \hat{a}_{jk} a_{j\ell}$$

Stability

Applying an ImEx method to

$$u'(t) = \underbrace{\lambda_1 u}_{\text{Explicit}} + \underbrace{\lambda_2 u}_{\text{Implicit}}$$

leads to

$$u_{n+1} = R(z_1, z_2) u_n.$$

So we can define the region of abs. stab:

$$S = \{(z_1, z_2) \in \mathbb{C}^2 : |R(z_1, z_2)| \le 1\}$$

This isn't as useful as it was for RK methods. Given

$$u'(t) = Au + Bu$$

can we prove that $|y|$ is bounded if

$$(h\lambda_1, h\lambda_2) \in S \quad \forall \; \lambda_1 \in \sigma(A), \; \lambda_2 \in \sigma(B) \, ?$$

Not in general! This only works if A and B are simultaneously diagonalizable. This can hold e.g. if we discretize a linear PDE spectrally or if one operator is diagonal (e.g.

$$u_t + a u_x = -u)$$

So it still may be useful.

Unfortunately, $\mathbb{C}^2$ is 4-dimensional, So it's challenging to characterize S.

Example: ImEx Euler

$$u_{n+1} = u_n + hf(u_n) + hL(u_{n+1})$$

$$R(z_1, z_2) = \frac{1+z_1}{1-z_2}$$

$$\left|\frac{1+z_1}{1-z_2}\right| \leq 1 \iff |1+z_1| \leq |1-z_2|$$

Clearly it holds if z_1, z_2 are in the stability regions of explicit and implicit Euler, respectively. But this isn't necessary.

A common approach is to study a 2D projection of S based on a particular equation. For instance, the advection-diffusion equation:

$$u_t + a u_x = b u_{xx}$$

Taking $u(x,t) = \hat{u}(t,\xi)e^{i\xi x}$ we have

$$\hat{u}_t = -i\xi a\hat{u} - \xi^2 b\hat{u}$$

So we take λ_1 imaginary and λ_2 negative real.

Exponential Methods

Formally, the exact solution of

$$u_t = f(u) + L(u)$$

with L linear is (Duhamel)

$$u(t+h) = e^{hL}u(t) + \int_0^h e^{(h-\tau)L} f(u(t+\tau))\,d\tau$$

We can't compute the integral exactly. If we approximate $f(u(t+\tau)) \approx f(u(t))$, we get

$$u_{n+1} = e^{hL}u_n + f(u_n)\int_0^h e^{(h-\tau)L}\,d\tau$$

$$= e^{hL}u_n + f(u_n)e^{hL}\int_0^h e^{-\tau L}\,d\tau$$

$$= e^{hL}u_n + f(u_n)e^{hL}\left[-L^{-1}e^{-\tau L}\right]_0^h$$

$$= e^{hL}u_n - e^{hL}L^{-1}(e^{-hL} - I)f(u_n)$$

$$= e^{hL}u_n + L^{-1}(e^{hL} - I)f(u_n)$$

$$= e^{hL}u_n + h\phi_1(hL)f(u_n) \qquad \phi_1 = \frac{e^z - 1}{z}$$

The general idea is to explicitly use the matrix exponential function. Higher-order methods can be constructed using a RK or multistep approach.

A major question is how to approximate the exponential, which can be tricky. Two leading methods are:

- rational (Padé) approximation
- Krylov subspace methods